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Research article

Physics-informed machine learning for specific capacitance prediction in supercapacitors using a curated experimental dataset and composite synergy analysis

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ABSTRACT

Supercapacitors are promising electrochemical energy-storage devices because of their high-power density and fast charge–discharge behavior. This work examines whether composite electrodes consistently achieve higher specific capacitance than pure-material electrodes and whether machine learning can explain the performance gains observed in a carefully curated experimental dataset. Unlike most previous studies, which primarily emphasize prediction accuracy, the present work focuses on developing a reliable experimental dataset for machine learning to enable more robust analysis and interpretation. A curated dataset comprising 910 experimental records from 52 peer-reviewed publications was developed, including carbon-based materials, metal oxides, metal sulfides, and composite electrodes. A total of 549 records with complete predictor variables were used for machine learning model development and evaluation. Pure carbon and metal oxide electrodes were retained as reference materials to enable direct within-study comparisons with composite systems. The dataset was curated using a no-assumption curation framework, physics-informed feature engineering, and DOI-stratified group splitting. CatBoost achieved an $R^2 = 0.948$ under random splitting, but performance dropped to $R^2 = 0.21$ under DOI-stratified group splitting. XGBoost, on the other hand, achieved the strongest cross-paper generalization, underscoring the importance of rigorous evaluation on literature-mined datasets. SHAP and ablation analyses identified the key features influencing specific capacitance prediction. Furthermore, Split Conformal Prediction was incorporated to provide calibrated uncertainty estimates. Together, these findings establish a reproducible benchmark for machine learning on literature-derived supercapacitor datasets.

1. Introduction

Supercapacitors are electrochemical energy-storage devices that store charge through Electrochemical Double-Layer Capacitance (EDLC) and Pseudo capacitance. They are ideally suited for a range of applications, from portable electronics and electric transportation to grid-scale energy storage, with their high-power density, fast charge-discharge capability, and long cycle life. However, accurate prediction of specific capacitance is still a challenge, as electrochemical performance is dependent not only on intrinsic materials chemistry but also on pore structure, morphology, electrolyte composition, current density, voltage window, and testing configuration. In experimental supercapacitor studies, X-ray diffraction [1,2] is commonly used to confirm phase purity, crystallinity, and composite formation, since these structural attributes directly influence conductivity, redox activity, and overall capacitance and recent Nd-doped $MnFeO_3$ and $AgNbO_3$/g-C_3N_4 electrodes provide representative examples.

Carbon-based systems are widely used among electrode materials due to their high conductivity, large accessible surface area, and good structural stability. However, surface ion adsorption often limits their charge storage. On the other hand, transition-metal compounds can provide a much larger pseudocapacitive contribution by means of

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reversible redox reactions, but they often suffer from lower conductivity and lower structural stability. This trade-off has resulted in the development of composite electrodes that combine carbon frameworks and redox-active metal compounds to benefit from both fast charge transport and additional electroactive sites [3,4]. A variety of capacitance values are reported in recent literature examples for pure carbons [5,6], metal oxides [7] and carbon–metal oxide composites [8,9], which suggest that composite architectures can outperform pure-material baselines under suitable conditions.

Machine learning (ML) is a promising approach for exploring structure-property relationships in such complex systems; however, current studies on supercapacitors have been limited mainly to narrow families of materials [10–19], such as carbon-based electrodes, porous carbons, CNT-based systems, $MgCo_2O_4$, conducting polymers, and MXene-derived materials. These studies, while achieving high predictive accuracy, are usually based on manually extracted literature datasets and evaluation through traditional random splits. Whereas general materials databases such as Materials Project and AFLOW provide computed descriptors rather than the experimental electrode-level variables for predicting capacitance, as a result, supercapacitor model development still relies on careful literature curation. Furthermore, differences in capacitance values measured using Cyclic Voltammetry (CV) and Galvanic Charge-Discharge (GCD) [14,15], incomplete reporting of experimental procedures, and paper-level data leakage could reduce the reliability and interpretability of published models [10–19].

To overcome these limitations, the present work develops a physics-informed machine-learning framework for specific-capacitance prediction based on a curated experimental dataset covering four major electrode classes: carbon materials, metal oxides, metal sulfides, and composite electrodes. In our work, we do not propose a new learning algorithm, but we stress methodological rigor utilizing careful dataset curation, preservation of missing-data semantics, electrochemically grounded feature engineering, mechanism-aware modeling, and DOI-stratified group-based evaluation. The framework is designed to test the hypothesis that composite electrodes exhibit synergistic performance compared to pure-material baselines and to provide a more realistic assessment of cross-paper generalization with heterogeneous literature-derived supercapacitor data [16].

1.1. Contributions and Novelty

- Curated an experimental dataset of 910 rows from 52 peer-reviewed papers, which includes carbon, metal oxides, metal sulfides, and composite electrodes, with pure-material baselines included by design.
- Directly tests the composite-synergy hypothesis by comparing composite electrodes with pure-material classes within a unified prediction framework.
- Preserving the missing-data semantics by separating not measured from not applicable with binary flag features, rather than collapsing all missingness into a single state.
- Introduced physics-informed electrochemical features including nonlinear current-density transforms, voltage-energy interaction terms, structural pore descriptors, and material-regime flags.
- Uses DOI-stratified group splits to separate within-paper interpolation from cross-paper generalization, revealing a large gap between random-split performance and grouped evaluation.
- Links model interpretation to electrochemical mechanisms by showing that ML-learned feature importances recover mechanism-dependent behavior across material classes.
- Extends deterministic prediction with Split Conformal Prediction, enabling calibrated intervals, empirical coverage assessment, and material-wise uncertainty analysis for screening decisions.

Section 2 reviews the recent machine learning techniques for supercapacitor capacitance prediction and highlights the methodological gaps. The dataset curation methodology is described in Section 3. Section 4 presents the composite synergy analysis on the curated dataset. We present exploratory data analysis in Section 5 to characterize the dataset and to identify key trends. Section 6 describes the physics-informed feature engineering strategy. Section 7 discusses the machine learning models. Section 8 presents the results of the feature interpretation. Uncertainty quantification is discussed in Section 9. This paper ends with Section 10 with main findings, limitations, and future research directions.

2. Related Work

In recent years, many researchers have shown interest in the application of machine learning techniques for predicting electrochemical performance in supercapacitors. Initially, studies used conventional regression and ensemble-learning techniques such as Random Forest, Support Vector Regression (SVR), and XGBoost to learn the relationship between the structural properties and specific capacitance, which achieved high R^2 values through train-test splitting. Most of the studies focused on small datasets, which are restricted to small categories of materials. The deep learning and neural network models achieved high prediction performance, while SHAP and Local Interpretable Model-agnostic Explanations (LIME) techniques were used to enhance the interpretability of the model predictions.

Despite these advances, several methodological limitations remain. Most studies are based on relatively small, material-specific datasets, frequently combine CV and GCD measurements, rely on conventional missing-data handling strategies, and evaluate models using random train–test splitting rather than group-aware protocols. Recent materials informatics studies have also explored graph neural networks [20,21] and transformer-based architectures [22], and large language models are used for property prediction and literature mining [23]. However, these approaches typically require large-scale or structure-rich datasets, whereas the present work focuses on a heterogeneous literature-derived experimental dataset. Here, some representative machine learning studies for the prediction of capacitance of supercapacitors are summarized in Table 1.

Although prior studies report high R^2 values, most of the research work is limited to single-material or closely related material families, such as carbon, graphene, CNTs, biochar, $MgCo_2O_4$, or activated carbon, which restricts their transferability to heterogeneous supercapacitor systems. Therefore, a new dataset is required to provide standardized, protocol-aware, and cross-material experimental data for more reliable and generalizable capacitance prediction.

3. Dataset Curation Methodology

The curated experimental dataset is described in terms of a systematic workflow for literature selection, data extraction, validation, and feature engineering. Fig. 1 provides an overview of the complete data-to-insights workflow, extending from dataset curation to machine learning model development, interpretation (SHAP and ablation analysis), and uncertainty quantification.

3.1. Literature Sourcing and Scope

Web of Science, Scopus, and Google Scholar were systematically searched to identify peer-reviewed studies on supercapacitor electrodes. To ensure dataset consistency, only three-electrode studies reporting specific capacitance from GCD were retained. Measurements at varying current densities were treated separately because capacitance is affected by the testing current. From an initial pool of >100 records, 52 studies published between 2012 and 2025 were selected [24–75] covering carbon materials, graphene/CNT composites, metal oxides, metal sulfides and hybrid pseudocapacitive systems, with pure carbon and metal

Table 1
Comparison with representative Machine Learning studies on supercapacitor capacitance prediction. *Group-split R: XGBoost scores 0.51 vs. CatBoost 0.21 under DOI-stratified evaluation.

References	Dataset size (rows)	Model	R^2	Key Findings	Limitations
Mishra et al. 2022 [10]	4899	XGBoost	0.93	Strong predictive performance achieved on carbon-based supercapacitor electrodes across diverse experimental conditions	Mixed testing methods; carbon-only dataset; no group split.
Liao et al. 2024 [11]	303	XGBoost	0.96	Nitrogen doping level and activation temperature identified as dominant predictors for biochar capacitance	Restricted to N-doped biochar materials; single dopant-material subclass
Tang et al.2024 [12]	300	RF / XGBoost	0.95	Synthesis temperature and metal ratio found to be primary performance drivers for $MgCo_2O_4$ electrodes	Single-material family ($MgCo_2O_4$); no group split
Tawfik et al. 2025 [13]	250	ANN	0.94	Functional group density and surface chemistry identified as key predictors for CNT-based electrode capacitance	CNT-only dataset; incomplete metadata; restricted material diversity
Ravichandran et al. 2024 [14]	CV dataset	Neural Network	0.90	Deep learning is effective for CV-base capacitance prediction; CV-trained models are not interchangeable with GCD-based models	CV behavior only; not interchangeable with GCD; no protocol separation
Saad et al. [15]	200	KNN/ANN	0.88	Nitrogen and oxygen atomic percentages were found to have a great effect on capacitance prediction.	Graphene-only morphology class; electrochemical protocol
Jamaluddin et al. [16]	167	MLP/RT /LR	0.81	MLP outperforms other algorithms; surface area and pore volume show the highest correlation with capacitance, with a correlation ratio of 0.4 each.	Biomass carbon only; KOH electrolyte only; precursors undifferentiated; no protocol separation; no group aware evaluation
Zhou et al. [17]	70	GLR/SVM/ RF/ANN	0.72	Mesopore surface area is identified as the dominant predictor at a high scan rate; increasing SSA beyond 1500 m^2/g does not improve capacitance. ANN best captures micropore-mesopore co-dependence.	Activated carbon EDLC only, 6 M KOH electrolyte only, CV-based data, only 70 data points, no group-aware evaluation.
Rahimi et al. [18]	101	ANN-RBF	0.99	Voltage window, electrolyte molarity and binder ratio identified as most important factors via sensitivity analysis; RBF-ANN with spread parameter 0.5 and 19 hidden neurons achieves lowest error	Biomass-derived activated carbon only, single material class, no composite or metal oxide coverage, no group-aware evaluation, likely overfitted on small dataset
Liu et al. 2024 [19]	275	LR /RF / XGBoost / LGBM	0.92	LGBM achieved the best predictive performance; SHAP analysis identified pore structure and heteroatom doping as dominant predictors for porous carbon supercapacitors	Restricted to N/O co-doped porous carbon materials; single electrolyte system (6 M KOH); no group-aware evaluation

oxides retained as benchmark references.

3.2. Core Extraction Philosophy

3.2.1. No-Assumption Principle

The dataset contains exclusive values directly reported in the original manuscripts, tables, and figures. No assumptions or estimations were made during the data extraction process. Missing parameters were labeled as 'not_measured' rather than being imputed.

3.2.2. Data Source Hierarchy

A source-priority hierarchy (TEXT > TABLE > GRAPH) was applied whenever the same parameter was reported in multiple formats. Values reported in the text or tables were retained because they were considered the most reliable. Graph values required digitization; they were assigned a medium confidence level.

3.2.3. Missing vs. Not-Applicable Taxonomy

A novel approach was developed by introducing flag columns with 3 states for missing data. The state measured was used for the properties that are measured and represented in the paper. The state "not_measured" was used for properties that can be measured but were not determined or reported, whereas the state "not_applicable" was for properties that are not applicable to a given material (e.g., metal loading for a carbon-only electrode or carbonization temperature for a metal oxide without carbon), and state measured was used for properties that are measured or determined or reported. This approach preserves the scientific meaning of missing values, unlike conventional imputation methods. During preprocessing, these categorical states were converted into binary indicator features.

3.3. Row-Level Data Structure

Each row represents a single experimental condition defined by the material, test mode, and operating condition. Material-level properties were shared across rows for the same material, whereas electrochemical testing parameters remained condition-specific, enabling prediction of specific capacitance as a function of material, structure, and testing condition.

3.4. Feature Categories

To preserve the physical meaning of experimentally reported supercapacitor behavior, the curated dataset was organized into five major feature categories. These categories include material features, synthesis features, structural features, composition features, and electrochemical features. This categorization reflects the multi-factor nature of capacitance generation.

3.4.1. Material Features

Material features characterize the broad identity of the material—for example, classification of the electrode system, precursor source, device configuration, biomass category, morphology-related indicators, and material type (carbon, metal oxide, sulfide, or composite). These features facilitate high-level circumstantial data regarding the most prominent charge-storage mechanism.

3.4.2. Synthesis Features

Synthesis features describe the chemical and thermal processing history of the electrode material, including carbonization temperature, activation agent, activation temperature, activation ratio, hydrothermal treatment, calcination temperature, annealing conditions, and synthesis atmosphere.

3.4.3. Structural Features

Structural features describe the intrinsic physical and pore-structure properties of the electrode material, including BET surface area, pore volume, average pore size, and Raman ID/IG ratio. Furthermore, surface chemistry, specifically the density and type of functional groups as analyzed via Fourier Transform Infrared (FTIR) spectroscopy, plays a critical role in enhancing charge-storage capacity [76]. However, quantitative FTIR descriptors were not included in the present dataset

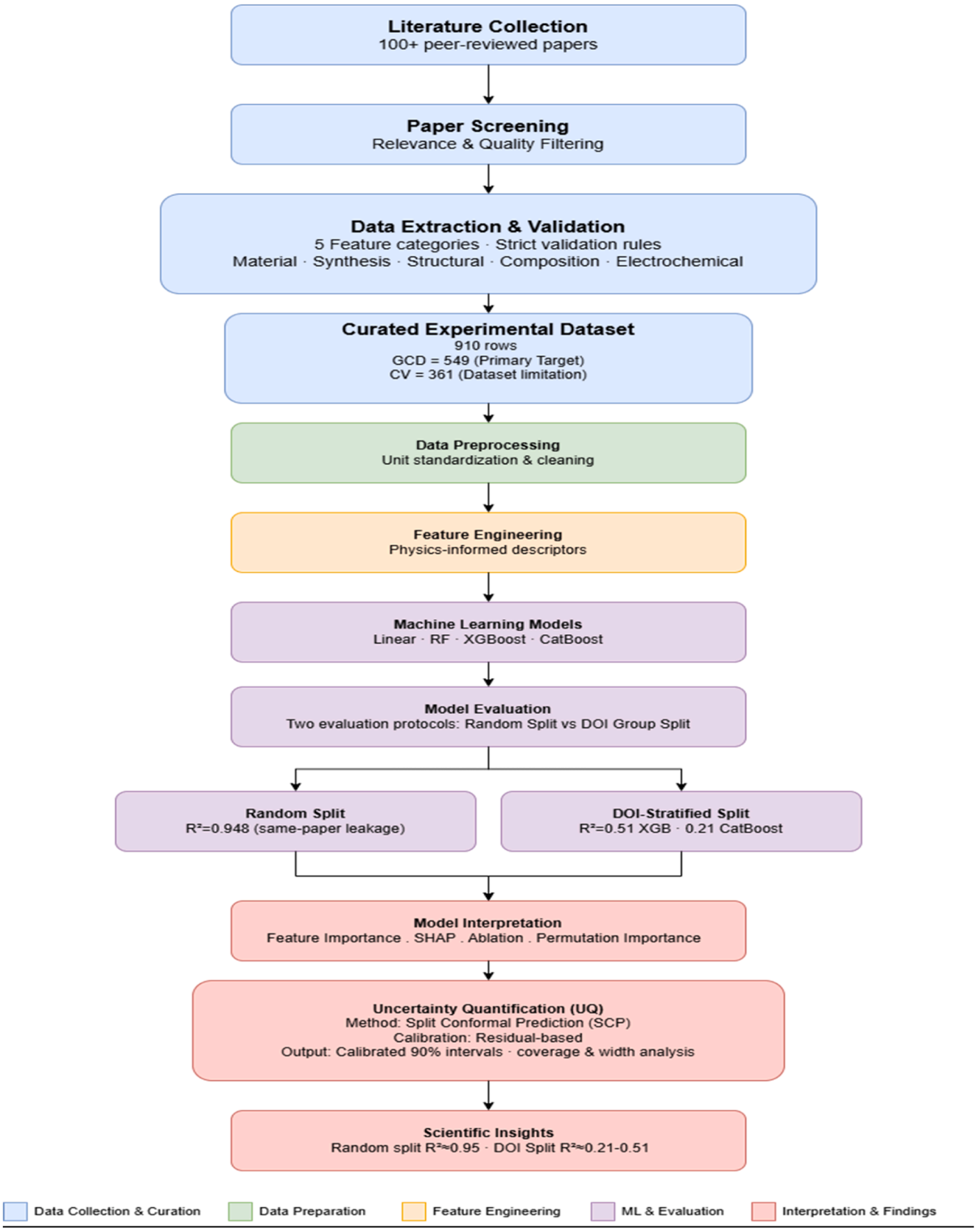


Fig. 1. Unified Data-to-Insights Pipeline: From Literature Curation to Model Interpretation.

because they were not consistently reported across the literature sources. Unlike electrochemical/testing features, structural properties are propagated across all rows that correspond to the same material because they are intrinsic material properties [77].

3.4.4. Composition Features

Composition features describe the chemically active constituents of the electrode material, including metal oxides, sulfides, dopants, heteroatom incorporation, loading percentage, and composite constituents.

3.4.5. Electrochemical Features

Electrochemical features characterize the experimental testing conditions and measured response variables connected with each observation, including current density, test mode, scan rate, potential window, electrolyte type, electrolyte concentration, power density, specific capacitance, Electrochemical Impedance Spectroscopy (EIS), and energy density parameters such as Rs and Rct [78]. Furthermore, the GCD analysis is widely used to assess the rate capability, capacitive nature, and conductivity, which verify the electrochemical performance of high-performance hybrid supercapacitor systems [79].

3.5. Prohibition-Based Validation

The dataset was curated using a predefined set of rules to ensure consistency and reproducibility. The values that are explicitly reported in the source literature were included, and inferred, extrapolated, or estimated values were excluded. Units were converted only when the original publication provided an explicit conversion. Duplicate records representing the same material under identical experimental conditions were removed. When the same parameter appeared in both graphical and textual or tabular form, the textual or tabular value was retained to avoid digitization errors.

3.6. Dataset Statistics

The final curated dataset contains 910 experimental rows from 52 studies with 199 unique electrode materials. Composite materials are the majority of the dataset with 624 rows (67.7%), followed by carbon with 168 rows (18.5%), metal oxide with 97 rows (11%), and metal sulfide with 14 rows (1.6%). Specific capacitance values span 0.8–3713.9 F/g; the mean value is 389 F/g, whereas the median is 240 F/g, reflecting the strong right skew of the dataset.

3.7. Preprocessing for Machine Learning Readiness

3.7.1. Experimental Space Restriction

The initial curated dataset comprised 910 experimental records retrieved from 52 independent works. The records were filtered using the following criteria: measurement_type = "electrode" (3-electrode cell), test_mode contained "GCD," current_density_unit was in ["A/g", "mA/g"], and capacitance_unit was in ["F/g", "mF/g"] to arrive at the final number of 549 records used for training and validating the machine learning model. The resulting subset of records was used to train the machine learning model on a consistent electrochemical protocol described by the majority of the included works. Though the final subset was considerably smaller than the initial dataset, the filtering is applied for focusing on the records describing similar experimental conditions and thus improving the validity of the subsequent analysis.

3.7.2. Unit standardization

Current density values reported in mA g^{-1} were converted to A g^{-1} and capacitance values reported in mF g^{-1} were converted to F g^{-1} to ensure numerical consistency is maintained across studies.

3.7.3. Numeric Range Handling

The qualifier is removed for the approximate values (e.g., '∼400'). Range-valued entries (e.g., '300–350 F/g′) are converted to their midpoint values (325).

3.7.4. Biomass Precursor Grouping

Raw precursor names are classified into six categories: agro_residue (silica-rich), shell_based (hard carbon), soft_biomass (leaves), protein_mineral (N/mineral-containing), woody_structural (lignin-rich), and food_waste (peels, seeds). Non-biological materials are defined as non-biomass.

3.7.5. Flag Feature Encoding

For each field with potential measured, not_measured, or not_applicable values, two binary features are generated: applicable and measured, as shown in Table 2. For each categorical property, binary indicator features were computed to preserve missing-data semantics during machine learning preprocessing.

4. Composite Synergy

This study evaluates whether composite electrode architectures deliver superior performance relative to their constituent materials. To enable direct comparative assessment under comparable experimental conditions, the curated dataset preserves within-study baselines comprising pure carbon, metal oxide, and metal sulphide electrodes alongside their corresponding composite systems.

4.1. Mechanism-Level Understanding

4.1.1. Pure Carbon Electrodes (EDLC)

Pure carbon electrodes store charge through electric double-layer capacitance (EDLC), which is based on the electrostatic adsorption of ions at the electrode–electrolyte interface without Faradaic reactions. Their performance depends largely on surface area, pore accessibility, and electrical conductivity. Consequently, they exhibit excellent cycling stability but relatively low specific capacitance.

4.1.2. Pure Metal Oxide/Sulfide Electrodes (Pseudocapacitance)

Metal oxides and sulfides store charge through reversible Faradaic redox reactions, giving them a much higher theoretical capacitance than EDLC materials. However, relatively low electrical conductivity and slow charge-transfer kinetics often limit their performance, particularly at high current densities.

4.1.3. Composite / Hybrid Electrodes (Synergistic)

Composite electrodes combine conductive carbon frameworks with pseudocapacitive metal oxides or sulfides to take advantage of both charge-storage mechanisms. The carbon framework improves electrical conductivity, facilitates ion transport, and provides structural support, while the metal oxide or sulfide contributes reversible redox-active sites. Together, these components improve charge transfer and electrochemical utilization, leading to higher specific capacitance than either component alone.

4.2. Quantitative Evidence of Composite Synergy

Composite electrodes achieved a mean specific capacitance of 451.2 F g^{-1} compared with 157.4 F g^{-1} for carbon electrodes, corresponding to a 187% improvement. Material-type indicators and

Table 2
Flag Feature Encoding Scheme Preserving Physical Meaning of Missing Data.

Original Flag	applicable	measured	Interpretation
measured	1	1	Property present and reported
not_measured	1	0	Present but unreported
not_applicable	0	0	Structurally inapplicable

interaction feature further enabled the machine-learning models to distinguish the performance regimes of EDLC, pseudocapacitive, and composite electrodes. This synergistic performance is illustrated in Fig. 2, which compares the specific capacitance and rate capability of composite electrodes against their pure-material counterparts.

Table 3 provides representative within-study comparisons showing that composite electrodes consistently outperform their corresponding carbon and metal oxide/hydroxide reference materials under comparable experimental conditions. These examples complement the corpus-wide statistical analysis and support the conclusion that composite architectures occupy a distinct high-performance regime within the curated dataset.

5. Exploratory Data Analysis

The preprocessing steps were validated by the means of Exploratory Data Analysis (EDA) that aims at identifying scientifically meaningful relationships within the data and potential model risks, such as imbalance, multicollinearity and non-linearity. The general idea of the EDA is to cast the prediction of supercapacitor capacitance as a non-linear, multi-factor and physics-driven regression problem.

5.1. Dataset Composition

Fig. 3 summarizes the material composition of the dataset. Composite electrodes account for 67.7% of all records, followed by carbon (18.5%), metal oxide (11.0%), and metal sulfide (1.7%) materials.

5.2. Target Distribution

Fig. 4 shows a strongly right-skewed capacitance distribution (median = 240 F g^{-1}; mean = 389 F g^{-1}), indicating that a relatively small number of high-performing electrodes dominate the upper tail of the dataset.

5.3. Performance by Material Type

Fig. 5 compares capacitance distributions across all the material classes. Composite electrodes exhibit the highest median and widest performance range, carbon electrodes being the lowest, and metal oxides exhibiting intermediate behavior. Although metal sulfides show high variability, they are represented by only 14 samples.

5.4. Biomass vs Non-Biomass Analysis

Fig. 6 shows that biomass-derived materials account for 45.9% of the dataset, with shell-based precursors representing the largest biomass category, followed by woody structural materials, agro-residues, and soft biomass. This distribution reflects the diversity of biomass feedstocks represented in the curated dataset.

5.5. Core Feature Scatter Plots

Fig. 7 shows weak individual relationships between most of the structural descriptors and specific capacitance values, whereas current density exhibits a clear inverse trend. These observations indicate that capacitance depends on non-linear interactions among multiple material and electrochemical variables rather than focusing on any single descriptor.

5.6. Comparative Statistics by Material Type

Table 4 summarizes capacitance statistics by material class. Composite electrodes achieve substantially higher mean and maximum capacitance than carbon electrodes, whereas metal oxides exhibit intermediate performance. Although metal sulfides show the highest mean capacitance, they are represented by only 14 observations.

5.7. Correlation Analysis

Fig. 8 shows that no individual feature exhibits a strong linear correlation with specific capacitance ($|r| < 0.5$). The presence of moderate inter-feature correlations and weak feature–target correlations support the use of nonlinear tree-based models for capacitance prediction.

6. Physics-Informed Feature Engineering

Physics-informed features were engineered to encode electrochemical relationships not explicitly represented by the measured variables, enabling the machine-learning models to capture physically meaningful nonlinear interactions, as summarized in Table 5, which depicts the engineered feature families used in this framework.

6.1. Current Density Transformations

The engineered features cd_log = log(1 + I) and cd_inverse = 1/ I capture the nonlinear dependence of capacitance on current density. These transformations represent the rapid decline in capacitance at low current densities and the diffusion-limited behavior observed at higher current densities.

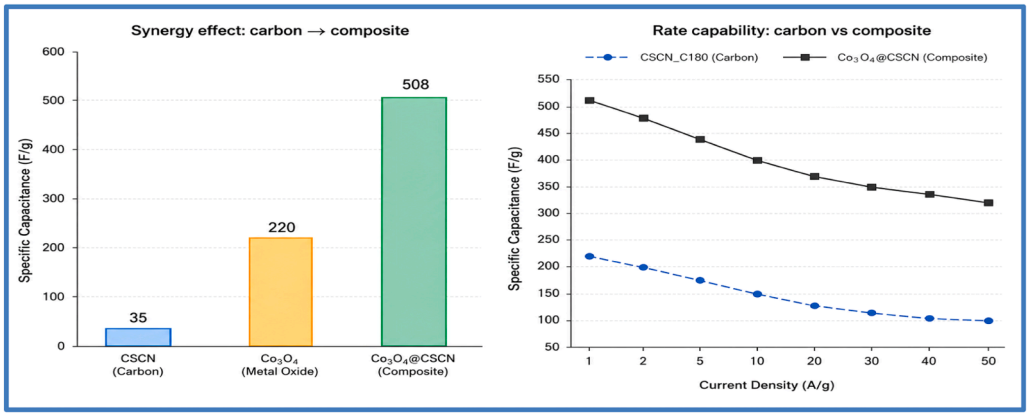


Fig. 2. Composite synergy demonstration. Left: Capacitance at 1 A/g for CSCN carbon (35 F/g), CSCN_C180 optimized carbon (220 F/g), and Co_3O_4@CSCN composite (508 F/g), showing ~1351% improvement from base carbon to composite. Right: Rate capability comparison confirming composite maintains advantage across all current densities [1,2,5,10,20,26,36,46]. Data from DOI 10.1002/bkcs.11389.

Table 3
Representative within-study comparisons demonstrating the synergistic performance of composite electrodes.

References	Carbon ($F\ g^{-1}$)	Metal Oxide/Hydroxide ($F\ g^{-1}$)	Composite ($F\ g^{-1}$)	Improvement vs Carbon (%)	Improvement vs Oxide/Hydroxide (%)
Chen et al. (2009) [24]	CNT (75)	V_2O_5 nanowire (146)	CVC−2 (313)	317.3	114.4
Fang et al. (2018) [25]	BPC (232.4)	N-Fe_2O_3 (128.3)	BPC/Fe_2O_3 (987.9)	325.1	670.0
Li et al. (2020) [26]	N-APC (111)	MnO_2 (200)	MnO_2@N-APC (330)	197.3	65
Boychuk et al. (2019) [27]	rGO (121)	$Ni(OH)_2$ (133)	$Ni(OH)_2$/rGO (494)	308.3	271.4

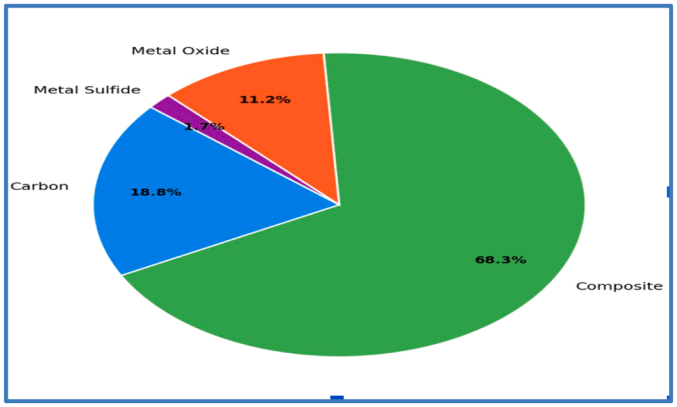


Fig. 3. Dataset composition by material type.

6.2. Voltage and Energy Features

Voltage-derived features capture the nonlinear dependence of stored energy on operating voltage and electrolyte environment through voltage-related transformations and voltage–electrolyte interactions.

6.3. Structural Interaction Features

The capacitance cannot be determined by a single structural parameter. To capture the co-dependence of surface area and pore architecture, three engineered features are introduced: surface_to_pore_ratio = SSA / pore volume describes pore compactness, pore_efficiency = pore volume / pore size represents effective volumetric pore distribution. ssa_pore_interaction = SSA × pore volume captures total accessible ion storage.

6.4. Graphitisation and Conductivity Features

The engineered feature structural_order_proxy = $1/I_D/I_G$ represents the degree of graphitisation such that higher values indicate more ordered, conductive carbon frameworks. Electrochemical resistance is described using these features: total_resistance = $R_s + R_{ct}$ and conductivity_index = 1/total_resistance. A wide dynamic range of resistance values is compressed using log-transformed variants.

6.5. Material Regime and Composition Flags

Binary indicators such as is_carbon, is_pseudocapacitive, is_metal_sulfide, is_composite, and is_metal_oxide are used to represent the dominant charge-storage mechanism. Regime-specific interaction features allow the model to learn the importance of the property depending on the material type, like pseudo_effect = conductivity_index × is_pseudocapacitive, ssa_carbon_effect = SSA × is_carbon, and ssa_composite_effect = SSA × is_composite.

6.6. Importance of Regime Features

The predictive contribution of regime-aware descriptors was evaluated through the DOI-grouped ablation analysis presented in Section 9.3.

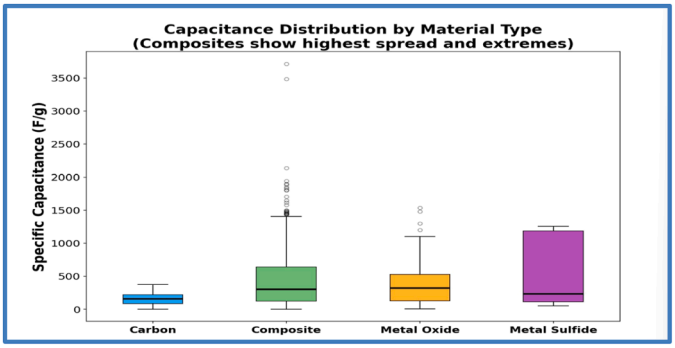


Fig. 5. Boxplot of specific capacitance by material type.

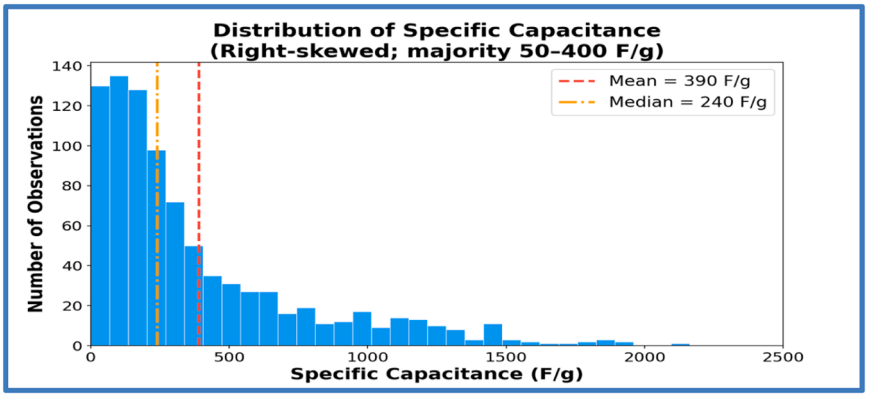


Fig. 4. Distribution of specific capacitance.

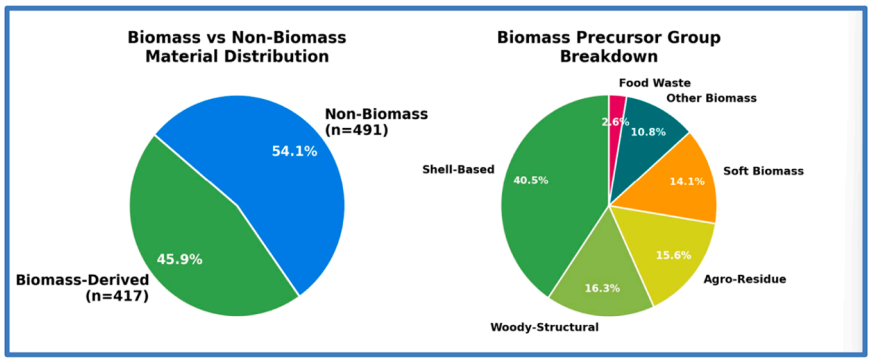


Fig. 6. Left: Biomass vs. non-biomass split in the dataset. Right: Biomass precursor group breakdown showing six chemically motivated categories.

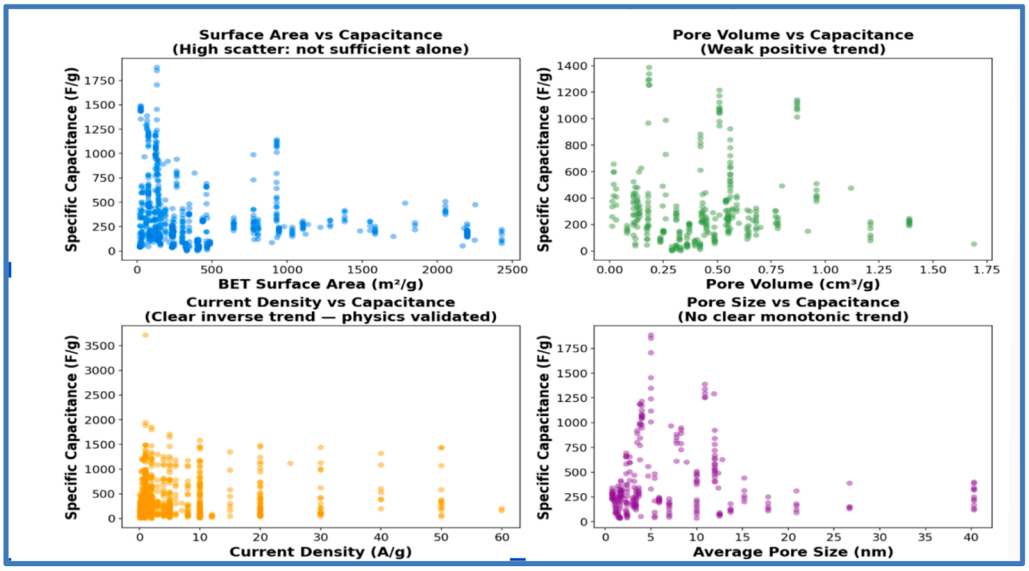


Fig. 7. EDA scatter plots of core features vs. specific capacitance.

Table 4
Descriptive statistics of specific capacitance stratified by material type.

Material Type	Min (F/g)	Max (F/g)	Mean (F/g)	Median (F/g)	Standard Deviation	No. of Samples
Carbon	0.8	375.0	157.4	160.0	83.2	168
Composite	1.0	3713.9	451.2	302.5	454.5	610
Metal Oxide	2.0	1536.0	372.5	322.7	328.6	97
Metal Sulfide	50.0	1257.0	614.2	231.1	542.5	14

7. Model Implementation and Evaluation

7.1. Model Selection

Following the feature engineering process explained in the previous section, four supervised machine learning algorithms were selected to fully evaluate the prediction relationship between material descriptors and specific capacitance. Random Forest and XGBoost were selected as representative tree-based ensemble methods that can capture complex nonlinear interactions between heterogeneous material features. The primary prediction model selected was CatBoost, due to its ordered boosting mechanism, built-in treatment of categorical variables without extensive preprocessing, and robustness to overfitting on structured tabular datasets. Two complementary data-splitting approaches were implemented to assess model performance in different validation settings.

7.1.1. Random Split Evaluation

The random split evaluation was performed using an 80:20 train–test partition, in which 80% of the dataset was used for model training and the remaining 20% was reserved for testing. Since the partitioning is performed randomly, samples originating from the same publication may be present in both subsets, allowing the models to be evaluated under conventional interpolation conditions. Under this evaluation strategy, CatBoost achieved the highest predictive performance with an R^2 score of 0.948, outperforming the other benchmark models and demonstrating its effectiveness in capturing the complex nonlinear relationships between physicochemical descriptors and the specific capacitance of carbon-based supercapacitor electrodes.

7.1.2. DOI Split Evaluation

Although the random split gives a good indication of how well the model can predict, it might overestimate the predictive power, since samples from the same publication might be present in both the training and testing sets. To overcome this limitation, we used a DOI-stratified group split with GroupShuffleSplit, ensuring that all samples from a publication are exclusively assigned to either the training or the test set. This rigorous test is based on predictions on previously unseen studies. As expected, the predictive performance decreased using this validation strategy, with XGBoost achieving the highest R^2 of 0.51 and CatBoost an R^2 of 0.21, showing the increased difficulty of cross-publication

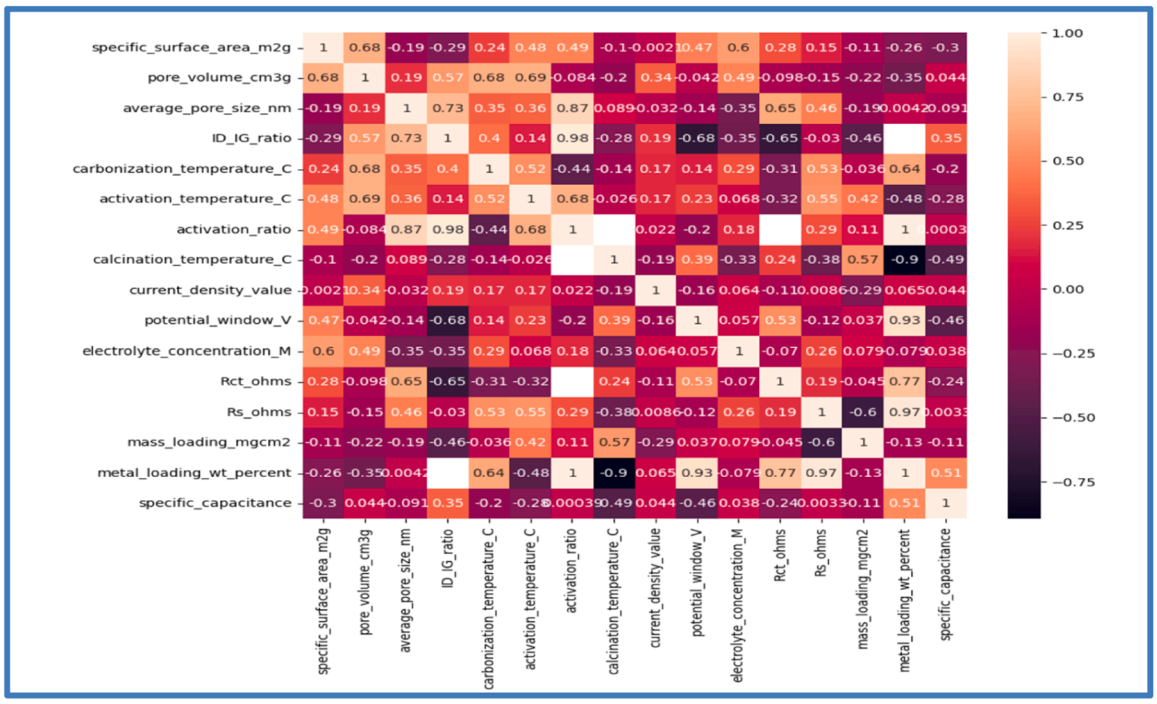


Fig. 8. Correlation heatmap demonstrating weak direct pairwise correlations between individual features and specific capacitance ($|r| < 0.5$ for all features).

Table 5
Feature space architecture showing the engineered feature families used in the machine-learning framework.

Feature Category	Representative Features	Role in the Framework
Missing-Value Flags	measured, applicable, available indicator variables	Preserve reporting information
Physics-Informed Features	voltage_squared, cd_log, cd_inverse, conductivity_index	Capture physical relationships
Regime & Interaction Features	is_carbon, is_composite, ssa_composite_effect, pseudo_effect	Mechanism-aware descriptors

generalization. The comparative performance of all evaluated models under the two validation strategies is discussed below and summarized in Table 6.

7.1.3. Comparative Performance and Model Selection

The comparative performance of the evaluated machine learning models under both validation strategies is summarized in Table 6. CatBoost was selected as the final model because it provided the best overall balance between predictive performance, native handling of categorical variables through ordered boosting, and robustness for structured tabular data. Accordingly, CatBoost was used for subsequent external validation, interpretability analysis, and uncertainty quantification.

Table 6
Model Performance under Random and DOI-Stratified Group Splits.

Model	R^2 (Random Split)	R^2 (DOI/ Group Split)	Interpretation
Random Forest	0.944	0.231	Strong within-paper performance but limited cross-paper generalization
XGBoost	0.943	0.514	Best cross-paper generalization
CatBoost	0.948	0.207	Best random-split performance, weaker cross-paper generalization

7.2. External Validation on an Independent Literature-Derived Dataset

In order to further assess the generalizability of the proposed framework, the final DOI-stratified CatBoost model was tested on an independent external validation dataset [80–84] which included 31 electrode-level GCD measurements not used for model development and hyperparameter optimization. Before predictions were made, the same preprocessing, feature engineering, and unit standardization pipeline was employed. The model was externally validated with R^2 of 0.313, RMSE of 447.60 Fg^{-1} and MAE of 343.31 Fg^{-1} . Predicted and experimental capacitance values correlated strongly linearly (Pearson $r = 0.871$, $p < 0.001$) and moderately monotonically (Spearman $\rho = 0.601$, $p < 0.001$), indicating that the model retained predictive power on unseen literature-derived samples despite the increased variability of the independent dataset.

8. Results and Interpretation

This section discusses the interpretation of the machine learning model with various analysis techniques. Feature importance, SHAP analysis, ablation studies and uncertainty quantification are used to investigate the behavior of the model, to validate the engineered features and to assess the reliability of the predictions.

8.1. Feature Importance Analysis

The contribution of each feature to the model predictions was measured by calculating the CatBoost feature importance. The resulting rankings confirmed the dominant role of the operating voltage in supercapacitor performance ($E \propto \frac{1}{2}CV^2$) as shown in Fig. 9. Potential_window_V and voltage_squared were the most influential predictors. Features related to precursor information, thermal processing, and engineered interaction terms also played a significant role, emphasizing the dependence of model predictions on both material properties and electrochemical operating conditions.

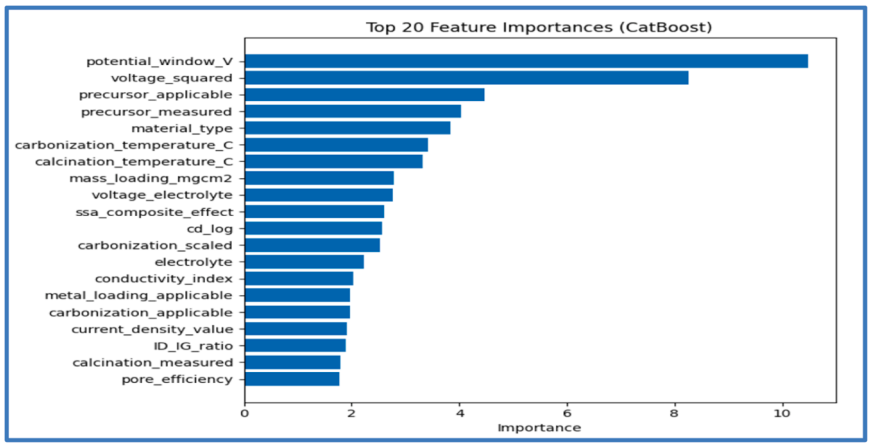


Fig. 9. Top 20 CatBoost feature importances highlighting voltage as the dominant predictor.

8.2. Permutation Importance

Permutation importance was calculated on the held-out DOI-grouped test set as a model-agnostic measure of global feature relevance to complement the CatBoost feature importance analysis. Feature importance was computed as the mean decrease in R^2 (ΔR^2) after random permutation of each feature over 30 repetitions, where higher positive values represented a greater contribution to predictive performance. The largest drop in predictive performance was generated by the voltage_squared, followed by carbonization_applicable, potential_window_V, is_composite, and is_carbon, as shown in Fig. 10. The rankings agree well with the CatBoost feature importance, suggesting the model relies mostly on physically meaningful descriptors related to voltage, material, and thermal processing. Features with near-zero importance values had little impact on predictive performance. The small negative values observed for some features are due to sampling variability during repeated permutations, not meaningful physical effects.

8.3. SHapley Additive exPlanations Analysis

8.3.1. Methodology and SHAP-confirmed Relationships

We used a TreeExplainer to perform SHAP analysis to measure the magnitude and direction of each feature's effect on individual predictions, shown in Fig. 11. SHAP, as opposed to global feature importance, shows if an increase in a feature increases/decreases the predicted specific capacitance, providing a physically interpretable view of model behavior. Table 7 summarizes the main SHAP relationships, where voltage-related variables, material type, biomass group, carbonization conditions, current density, conductivity, mass loading, and engineered interaction features demonstrate electrochemically consistent trends.

Collectively, the SHAP results demonstrate that the model captures electrochemically meaningful relationships rather than paper-specific numerical patterns. The consistency observed across nine independently evaluated features–response relationships provides additional confidence in the physical interpretability of the learned model.

Although the SHAP analysis shows that the final model captures physically meaningful electrochemical relationships, the SHAP analysis does not quantify the contribution of the engineered feature groups introduced during feature construction. Hence, an ablation study was performed to evaluate the additive value of each feature group.

8.4. Ablation Study

8.4.1. Experimental design and statistical analysis

We quantified the incremental contribution of the engineered feature groups through a sequential ablation analysis comprising six feature sets (shown in Table 8): (i) base descriptors, (ii) base + physics-informed

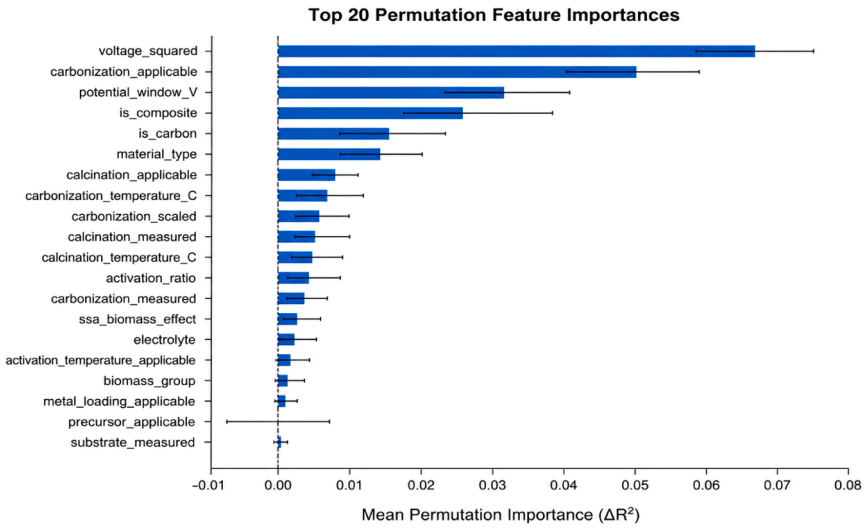


Fig. 10. Top 20 permutation feature importances for the CatBoost model evaluated on the held-out DOI-grouped test set. Feature importance is expressed as the mean decrease in the coefficient of determination (ΔR^2) over 30 permutation repeats. Error bars represent one standard deviation.

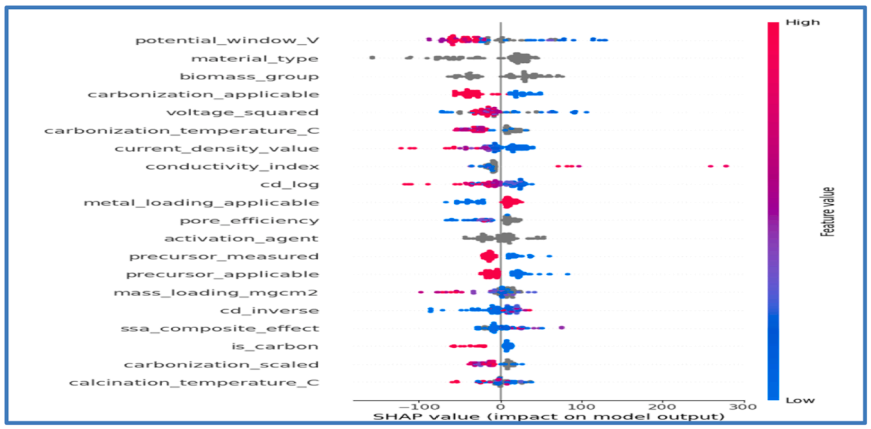


Fig. 11. SHAP beeswarm analysis of the CatBoost model demonstrating feature effects on capacitance predictions.

Table 7
SHAP-Based Analysis of Feature Importance and Electrochemical Mechanisms Regulating Capacitance.

Feature	SHAP direction	Electrochemical Rationale
potential_window_V ↑	Capacitance ↑	E ∝ ½CV²; wider window increases energy storage ceiling
voltage_squared ↑	Capacitance ↑	Quadratic amplification of window effect
current_density_value ↑	Capacitance ↓	Rate-limited ion diffusion restricts accessible surfaces.
conductivity_index ↑	Capacitance ↑	Lower resistance enables fast electron transport to active sites
mass_loading_mgcm2 ↑	Capacitance ↓	Thick electrodes create diffusion-length limitations.
is_composite = 1	Capacitance ↑	Dual EDLC + pseudocapacitive mechanism activation
carbonization_temp (700–900°C)	Capacitance ↑	High SSA amplifies both charge-storage mechanisms in hybrids
cd_inverse ↑	Capacitance ↑	Slow rate → full pore penetration → maximum accessible interface
ssa_composite_effect ↑	Capacitance ↑	High SSA amplifies both charge-storage mechanisms in hybrids

features, (iii) base + physics-informed + electrochemical-informed features, (iv) base + physics-informed + electrochemical-informed + interaction-informed features, (v) base + physics-informed + electrochemical-informed + interaction-informed + regime descriptors, and (vi) the full model. At each stage, CatBoost was trained with identical hyperparameters and evaluated over 30 independent DOI-grouped GroupShuffleSplit (80:20) repetitions. Mean R^2, RMSE, and MAE, together with their 95% confidence intervals, were computed across repeated splits. Pairwise comparisons between consecutive stages were initially estimated using paired tests on split-wise performance score because the study involves repeated resampling and multiple sequential comparisons. The inferential results are interpreted conservatively, emphasizing the magnitude and consistency of the stage-wise changes rather than a binary significance label alone.

8.4.2. *Stage-by-stage interpretation*

Under careful DOI-grouped evaluation, all ablation stages resulted in negative or near-zero mean R^2 values, indicating that cross-paper prediction is much harder than within-paper interpolation. Mean R^2 improved from −0.077 to −0.030 with inclusion of physics-informed descriptors and RMSE and MAE reduced slightly. The enhancement was modest and not confirmed after multiplicity correction. No reproducible incremental benefit was observed when using electrochemical informed descriptors. We also found that explicit interaction features did not consistently improve generalization, indicating that CatBoost was already able to capture much of the relevant nonlinear structure without the need for additional manually engineered interaction terms. The largest numerical improvement was observed when regime descriptors

Table 8
DOI-grouped ablation analysis of sequential engineered feature groups for CatBoost performance prediction.

Stage	Mean R^2 (95% CI)	Mean RMSE (95% CI)	Mean MAE (95% CI)	Raw p-value vs. previous stage	Holm-adjusted p-value	Interpretation
Base	−0.077 (−0.261, 0.107)	392.18 (364.84, 419.52)	299.10 (278.26, 319.94)	–	–	Baseline DOI-grouped model
Base + Physics-informed features	−0.030 (−0.161, 0.101)	389.72 (360.08, 419.35)	290.85 (269.27, 312.44)	0.166	0.664	Small numerical gain; no reliable evidence of improvement
Base + Physics-informed + Electrochemical-informed features	−0.031 (−0.209, 0.146)	380.77 (355.02, 406.52)	289.81 (269.98, 309.65)	0.967	1.000	No evidence of additional improvement
Base + Physics-informed + Electrochemical-informed + Interaction-informed features	−0.051 (−0.242, 0.140)	384.30 (359.10, 409.51)	292.39 (273.51, 311.27)	0.284	0.852	No evidence of additional improvement
Base + Physics-informed + Electrochemical-informed + Interaction-informed + Regime descriptors	−0.0002 (−0.180, 0.179)	374.73 (347.11, 402.35)	279.15 (255.35, 302.95)	0.032	0.160	Largest numerical gain. Suggestive, but not significant after multiplicity correction
Full model	−0.006 (−0.169, 0.157)	378.33 (350.12, 406.53)	278.30 (252.87, 303.72)	0.756	1.000	No additional gain beyond regime descriptors

were introduced, reducing mean RMSE from 384.30 to 374.73 F g^{-1} and mean MAE from 292.39 to 279.15 F g^{-1} . However, although this stage yielded the lowest raw p-value, the effect did not remain below the 0.05 threshold after Holm correction. Extending the regime-stage model to the full feature set did not produce any additional improvement. Overall, the ablation study provides quantitative but cautious support for mechanism-aware regime descriptors as the most informative engineered feature family under DOI-grouped evaluation, whereas the remaining feature groups show limited and statistically inconclusive incremental benefit.

8.5. Uncertainty Quantification

The DOI-based grouping adopted for model evaluation provides a rigorous assessment of the model's ability to generalize to previously unseen publications. However, this evaluation strategy does not quantify the uncertainty or confidence associated with individual predictions. In practical materials informatics applications, predictive accuracy alone is insufficient, as experimental decision-making also requires an assessment of prediction reliability and uncertainty. Therefore, uncertainty quantification was performed using Split Conformal Prediction (SCP) to generate statistically calibrated prediction intervals without modifying the underlying CatBoost model or the trained regression framework. SCP provides calibrated prediction intervals based on residuals on the calibration set, thus complementing the analysis of generalization with statistically meaningful confidence estimates, as opposed to the usual regression metrics that only give point predictions.

Table 9 presents the overall uncertainty performance of the proposed framework. In the random-split evaluation, SCP achieved an empirical coverage of 92.7%, very close to the 90% nominal confidence level, with an average width of 572.7 Fg^{-1} for the prediction intervals. This suggests that the confidence estimates are well calibrated and useful in practice. However, the DOI-grouped evaluation obtained an empirical coverage of 76.8% with a larger mean prediction interval of 1278.1 F g^{-1} . This larger interval shows the much greater uncertainty associated with predicting entirely new publications. These results suggest that uncertainty estimation is a relevant strategy for adapting to the increased distributional shift observed in cross-publication evaluation, thus complementing the generalization analysis.

Overall, Table 9 depicts that uncertainty calibration was satisfactory under the random split but faced degradation under the DOI-stratified split.

To further assess the robustness of the uncertainty estimates, empirical coverage was studied across different material classes, as summarized in Table 10. The empirical coverages of carbon (96.3%) and composite (91.7%) electrodes, which constituted the majority of the evaluation dataset, were close to the target confidence level (88.9%) for metal oxides. For metal sulfides, the coverage has been 100% empirically, but that must be taken with a grain of salt, as only two test samples were used. For standard Split Conformal Prediction, the width of the prediction interval was the same for all material classes, as only one global conformal quantile was used, while the coverage differed depending on the underlying experimental variability and sample distribution.

Interval width remained constant across all material classes because standard Split Conformal Prediction applies a single global conformal quantile to every prediction. This quantile was computed from the sorted absolute residuals of the held-out calibration set using the finite-sample 90% conformal procedure, resulting in fixed interval widths of 572.74 F/g for the random split and 1278.1 F/g for the DOI-stratified split, while empirical coverage varied across material classes. Uncertainty quantification had better coverage in the random splitting than in the DOI-stratified evaluation, consistent with the higher cross-study heterogeneity of the DOI-based test set. Fig. 12 presents the prediction interval parity plots for both evaluation strategies, where the vertical bars denote the conformal prediction intervals and the dashed diagonal represents perfect prediction. Samples whose intervals intersect the diagonal are correctly covered, whereas those lying entirely above or below it correspond to coverage misses, consistent with the coverage statistics reported in Table 9.In practical terms, the prediction intervals gives a quantifiable confidence measure for each predicted capacitance. For example, a predicted 520 F/g for the random-split calibration is interpreted as 520 $\pm$ 287 F/g, representing the probable range of the true capacitance with 90% confidence. This uncertainty information enables researchers to assess not only how well a candidate material is predicted to perform, but also how reliable that prediction is, allowing for better-informed prioritization of materials for experimental validation.

Table 9
Coverage and interval width, Split Conformal Prediction.

Metric	Random Split	DOI Split
Nominal Coverage (%)	90.0	90.0
Observed Coverage (%)	92.7	76.8
Coverage Error (pp)	+2.7	−13.2
Average Interval Width (F/g)	572.7	1278.1

Table 10
Empirical coverage across material classes under the random-split evaluation.

Material Class	Number of test samples of each material class	Empirical Coverage (%)
Carbon	27	96.3
Composite	72	91.7
Metal Oxide	9	88.9
Metal Sulfide	2	100.0

9. Discussion

9.1. Significance of the Curation Methodology

This curation methodology provides a reproducible way to build machine-learning-ready experimental datasets while preserving the scientific meaning of reported measurements by explicitly distinguishing missing and non-applicable variables. The lower predictive performance under DOI-stratified group evaluation indicates that cross-study distribution shifts remain a key challenge for literature-derived datasets and highlights the need for rigorous evaluation.

9.2. Significance of the Composite Synergy Finding

The curated dataset consistently shows that hybrid electrode architectures perform better than their individual component materials when evaluated under comparable experimental conditions. Because composite and pure-material electrodes were compared within individual studies, the results are less affected by differences in experimental conditions between studies. The machine-learning models consistently reproduced these relationships, providing additional confidence in the reliability of the curated dataset.

9.3. Significance of Random and DOI split

The performance gap between random-split and DOI-grouped evaluation provides valuable insight into the robustness of the proposed framework. Random splits might overestimate predictive performance, because materials from the same publication could have similar experimental conditions and characteristics, enabling the learning of publication-specific information. In contrast, DOI-grouped evaluation tests the model on completely unseen studies, which is a more realistic measure of cross-publication generalization. The results emphasize the

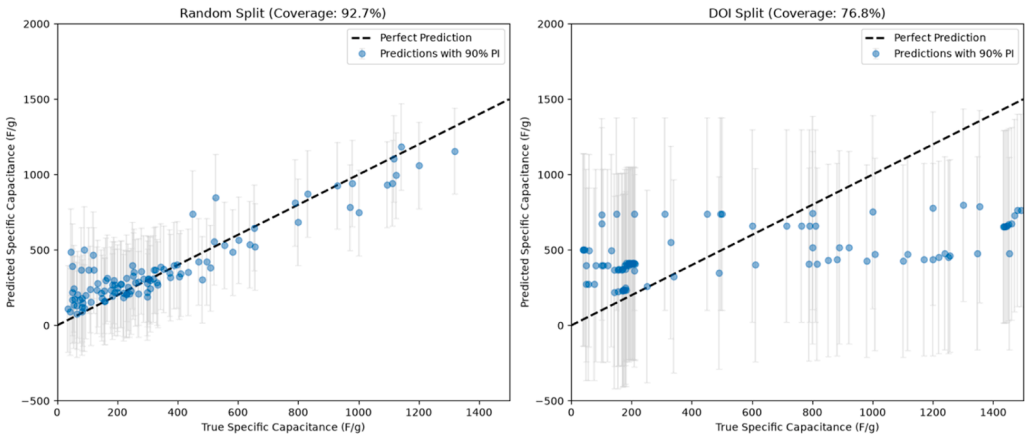


Fig. 12. Prediction Interval Parity plot.

necessity of validation at the publication level to create reliable and practically deployable materials informatics models.

9.4. Dataset Scope and Representativeness

Although the curated corpus contains 910 experimental records from 52 independent studies, the machine-learning models were developed using a filtered GCD-only subset of 549 experimentally comparable samples after applying the consistency criteria described in Section 5.1. The modeling dataset comprises 129 unique material formulations from 38 studies, with strong representation of aqueous electrolytes (predominantly KOH and Na_2SO_4), composite electrodes, etc. Consequently, the learned relationships and DOI-grouped model performance should be interpreted within the experimental domain represented by the current dataset. Future expansion through additional experimentally validated studies, broader material systems, and more diverse electrochemical testing conditions is expected to improve model robustness and generalisability.

Although this study focuses on predicting specific capacitance, the curated dataset also contains additional electrochemical performance metrics, including rate retention (392/910), capacitance retention (200/910), cycle life (197/910), energy density (76/910), and power density (75/910). However, the substantially lower coverage of these variables currently limits their suitability for reliable predictive modeling. As larger and more comprehensive datasets become available, the proposed framework can be extended to simultaneously predict multiple electrochemical performance metrics.

The proposed framework for uncertainty quantification yields calibrated prediction intervals with finite-sample guarantees. Nonetheless, the fairly wide prediction intervals, particularly in the DOI-stratified evaluation, indicate that the confidence in prediction diminishes when the model is applied to diverse literature-derived datasets. Future work will increase the dataset with more diverse experimental studies and material systems to improve the calibration and the predictive uncertainty.

10. Conclusion & Future Scope

This work presents one of the first systematically curated, electrode-level, GCD-focused supercapacitor datasets confirming that composite and hybrid electrodes substantially outperform carbon-only electrodes, with a mean specific capacitance of 451 F/g versus 157 F/g (+187%). A strict no-assumption curation strategy incorporating explicit missing-value taxonomy, chemistry-informed categorization, physics-informed feature engineering, and DOI-stratified group evaluation improved the reliability and interpretability of machine learning models. Repeated DOI-grouped ablation identified regime/material behavior descriptors as the only statistically significant contributors to cross-paper predictive performance. Although CatBoost achieved $R^2 = 0.948$ under random splitting, performance declined to $R^2 = 0.21$ under DOI-stratified evaluation, demonstrating that much of the apparent accuracy under random splitting reflects paper-specific reporting patterns rather than transferable material physics.

SHAP analysis identified voltage window, current density, and conductivity as the dominant predictors. Together, these findings establish a reproducible benchmark for machine learning on literature-derived supercapacitor datasets and demonstrate that cross-paper generalization, rather than within-paper interpolation, is the appropriate measure of model utility. Consequently, the corpus-wide composite-synergy effect and the material-intrinsic feature rankings should be interpreted as experimentally testable hypotheses for future investigation rather than validated design recommendations or prescriptive synthesis rules. The trained CatBoost model was evaluated for the reliability of individual predictions using Split Conformal Prediction. However, uncertainty quantification provides a calibrated prediction interval for each prediction and allows the confidence associated with the predicted specific capacitance to be assessed, as opposed to traditional regression metrics.

In future the dataset can be extended by including the CV-derived capacitance with strict protocol separation and scan-rate-normalized targets, and other electrochemical metrics such as rate retention, energy density and power density. Descriptors from DFT, transfer learning and domain adaptation could improve generalization across papers. With larger and more comprehensive datasets, we could explore graph neural networks, transformer-based models, and large language model-assisted literature mining.

Financial Interest

The authors have no relevant financial or non-financial interests to disclose.

Data & Code Availability

The dataset generated and analyzed during this study will be made publicly available on Kaggle upon publication. The source code used for data preprocessing, feature engineering, machine learning model development, and analysis will be made publicly available through the GitHub repository (https://github.com/khub-crystalparadigm/capacitance_pred).

Authors' contribution statements

Shilpa Choudhary and Kasarapu Venkataramana supervised the overall research work, conceptualized the study, guided the

methodology, and reviewed the manuscript. Sreethi Balasa, Kaushik Pashupathi, S. Pramod Reddy, and Akshith Ponugoti contributed to data collection, experimentation, implementation, analysis, and preparation of the initial manuscript draft as research interns. All authors reviewed and approved the final version of the manuscript.

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CRediT authorship contribution statement

Kasarapu Venkataramana: Writing – review & editing, Supervision, Conceptualization, Project administration. **Sreethi Balasa:** Writing – original draft, Software, Formal analysis, Data curation. **Kaushik Pashupathi:** Writing – original draft, Software, Formal analysis, Data curation. **Pramod Reddy S:** Writing – original draft, Software, Formal analysis, Data curation. **Shilpa Choudhary:** Writing – review & editing, Supervision, Methodology, Conceptualization, Project administration. **Akshith Ponugoti:** Writing – original draft, Software, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Dataset Curation Rules (Supplementary)

Table A1
Complete extraction rulebook for reproducibility

Rule Category	Rule	Rationale
No-assumption	ONLY extract explicitly reported values; never infer, interpolate, or back calculate	Prevents systematic bias propagation
Missing taxonomy	Distinguish not_measured (exists, unreported) from not_applicable (structurally absent)	Preserves scientific meaning of absence
Data source hierarchy	Text > Table > Graph; remove graph value if text/table found	prioritizes semantic certainty over visual estimation
Protocol separation	Never mix CV and GCD rows; bind capacitance always to its condition	Prevents protocol-induced bias
Unit conversion	Only convert with explicit author-stated equivalence; mA/g → A/g, mF/g → F/g	Prevents fake unit-driven correlations
No duplication	No duplicate rows for same (material, condition) pair	Prevents model overfitting to repeated data
Graph confidence	Mark graph-derived values: data_source=graph, confidence=medium (±5–15% error)	Preserves uncertainty information
Feature consistency	Material-level features identical across all rows for same material	Ensures model learns condition-dependent capacitance

.

Appendix B. Full Feature List

Table B1
Full feature list with category, type, unit, missingness, and source

Feature Name	Category	Type	Unit	Missingness	Source
material_type	Material	Cat	—	Low	Extracted
biomass_group	Material	Cat	—	Low	Engineered
electrode_configuration	Material	Cat	—	Low	Extracted
carbonization_temp_C	Synthesis	Num	°C	High	Extracted
activation_temperature_C	Synthesis	Num	°C	High	Extracted
activation_ratio	Synthesis	Num	ratio	High	Extracted
specific_surface_area_m2g	Structural	Num	m^2/g	Moderate	Extracted
pore_volume_cm3g	Structural	Num	cm^3/g	Moderate	Extracted
average_pore_size_nm	Structural	Num	nm	High	Extracted
ID_IG_ratio	Structural	Num	—	High	Extracted
current_density_value	Electrochemical	Num	A/g	Low	Extracted
potential_window_V	Electrochemical	Num	V	Moderate	Extracted
electrolyte	Electrochemical	Cat	—	Low	Extracted
Rs_ohms / Rct_ohms	Electrochemical	Num	Ω	High	Extracted
{field}_applicable	Flag	Binary	0/1	None	Engineered
{field}_measured	Flag	Binary	0/1	None	Engineered
cd_log, cd_inverse	Physics	Num	—	Low	Engineered
voltage_squared	Physics	Num	V^2	Moderate	Engineered
conductivity_index	Physics	Num	Ω^{-1}	High	Engineered
ssa_pore_interaction	Physics	Num	$m^2 \cdot cm^3/g^2$	High	Engineered
is_carbon / is_composite / is_metal_oxide / is_metal_sulfide	Regime	Binary	0/1	None	Engineered
ssa_carbon_effect / ssa_composite_effect	Regime	Num	m^2/g	High	Engineered
pseudo_effect	Regime	Num	Ω^{-1}	High	Engineered

.

Appendix C. Hyperparameter Tuning Details

Table C1
CatBoost hyperparameter search space and final configuration

Hyperparameter	Search Range	Final Value	Justification
iterations	500 – 5000	3000	Sufficient convergence without excessive runtime
learning_rate	0.01 – 0.1	0.03	Slow learning reduces overfitting on moderate dataset
depth	4 – 8	6	Captures multi-feature interactions; 8 begins to overfit
l2_leaf_reg	1 – 10	5	Regularisation stabilises leaf weights on sparse features
bagging_temperature	0 – 1	1	Maximum bootstrapping randomness improves robustness
early_stopping_rounds	50 – 200	100	Stops training when validation R^2 plateaus

.

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